Data Analytics Capstone

Learning When to Repeat and When to Explore: Constraint-Aware Auto-Ordering for School Meal Subscriptions

Interim Report

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GitHub Repository

Code, notebooks, documentation, and data files for this project are available at:

https://github.com/kkpathak91/school-meal-autoorder

The complete raw dataset is attached to the repository's release v1.0:

https://github.com/kkpathak91/school-meal-autoorder/releases/tag/v1.0

The repository contains the data generator and validation scripts (*src/*), the analysis notebooks (*notebooks/*), the processed student-level feature file (*data/processed/*), and the data card (*docs/*). Its structure is shown in Figure 2, and instructions for reproducing every result are in the repository's README file and in Appendix C.

Table of Contents

Introduction 5

Background and Context 5

Problem Statement 5

Purpose of the Study 6

Interim Project Status (Progress Snapshot) 6

Scope and Objectives 6

Literature Survey 7

Literature Review Approach 7

Summary of Key Literature 8

Data Description 11

Data Source(s) and Access 11

Dataset Overview 12

Data Dictionary (Mandatory) 13

Sample Size Determination 14

GitHub Data Availability Statement 16

Screenshots / Evidence 16

Analysis 17

Data Cleaning 17

EDA Results (Interpretation Emphasis) 19

Modelling 25

Choice of Models with Justification 25

Features Included and Feature Engineering 26

Evaluation Metrics (Include Formulae and Calculations) 28

Preliminary Results 30

Preliminary Findings by Research Question 30

Segmentation Results 33

Interim Limitations and Risks 33

Next Steps (for Final Report) 34

Bibliography 35

Appendix A: Feature Screening and Segment Selection 38

Appendix B: Statistical Tests Used 40

Appendix C: Reproducible Workflow 40

Introduction

Background and Context

Many private and public schools in the United States outsource lunch to caterers that run online pre-ordering platforms. Each day a caterer publishes a menu of six to ten entrées for every school, and families choose their child's lunch in advance, either one meal at a time or through a weekly subscription plan that covers fixed weekdays and a fixed number of side items. The stakeholders are the families who order, the students who eat, the schools that host the service, and the caterer, whose kitchens and delivery routes depend on accurate orders.

Manual ordering creates order fatigue: parents must repeat similar choices every week, and lunches are missed when they forget. Caterers therefore want to place lunches automatically for subscribers. An automatic choice only creates value if the family keeps it, so a poor pick leads to a cancellation, a swap, or a wasted meal. Plate waste in US school meal programmes is already substantial (Byker Shanks et al., 2017), so lunches that children actually want also matter beyond the business case.

The setting is analytically distinctive. Children repeat favourite items heavily, which the recommender-systems literature calls repeat consumption (Ren et al., 2019; Li et al., 2023). The set of choices is small, fixed by a published menu, and rotated on a schedule, which is a form of concept drift (Gama et al., 2014). Every recommendation must also respect hard constraints: the day's menu, the child's allergens and dietary preferences, and the subscription plan.

Problem Statement

A legacy auto-ordering rule at the study's provider places each subscriber's usual item for that weekday. In the study data, these auto-orders are cancelled at 24.8%, against 9.2% for lunches that families order themselves, which suggests the rule often picks something the child does not want. The rule ignores three things: how much variety each child prefers, the regular rotation of menus, and the lack of history for new students. No published work addresses personalised, constraint-bound meal ordering for children, so it is not known how much better a learned approach can do, where it fails, or which students it serves least well. This project

addresses that gap.

Purpose of the Study

The study aims to **predict**, for each subscriber and each plan day, which of that day's eligible entrées the student would choose, by **ranking** the candidate items. It also **segments** students by ordering behaviour to explain where predictions succeed or fail, **classifies** legacy auto-orders by their risk of cancellation, and **explains the drivers** of both outcomes through feature importance. The practical goal is an evidence-based design for an auto-ordering system that families keep.

Interim Project Status (Progress Snapshot)

- **Completed:** data acquisition and validation; data cleaning (missing values, duplicates, outliers); univariate and bivariate exploratory data analysis (EDA); hypothesis-driven EDA with baselines for all four research questions (RQs); feature engineering and screening; behavioural segmentation with k-means; sample-size verification; preliminary models for RQ1, RQ2, and RQ4 (logistic regression and Light Gradient Boosting Machine [LightGBM]).
- **In progress:** variety-calibrated re-ranking for RQ1; item-substitution features for RQ2; hyperparameter tuning of the LightGBM ranker.
- **Pending:** model-based cold-start comparison for RQ3; mixed-effects testing for RQ2; robustness checks on a second data seed and an external public dataset; final validation and recommendations.

Scope and Objectives

The research problem is to predict a child's lunch choice under hard constraints well enough that families keep automatic orders. Four research questions structure the work (Table 1); for each, *h* denotes Hit@1 (defined under Evaluation Metrics) and *c* the cancellation rate.

Table 1

Research Questions and Hypotheses

RQ	Research question	Null hypothesis (H_0)	Alternative (H_1)
RQ1	Does a learned ranking model predict the chosen entrée better than the strongest rule-based baseline?	$h_{\text{model}} = h_{\text{baseline}}$	$h_{\text{model}} > h_{\text{baseline}}$
RQ2	Does prediction accuracy fall in the ten school days after a menu-cycle change?	$h_{\text{post-change}} = h_{\text{mid-cycle}}$	$h_{\text{post-change}} < h_{\text{mid-cycle}}$
RQ3	After how many orders does a student's own history predict better than a cohort guess?	Personal Hit@1 never exceeds cohort Hit@1	Personal Hit@1 exceeds cohort Hit@1 after k orders
RQ4	Are auto-orders cancelled more when the pick does not match the student's preference, and can cancellation be predicted?	$c_{\text{mismatch}} = c_{\text{match}}$	$c_{\text{mismatch}} > c_{\text{match}}$

In scope: entrée prediction for students on plan days from 14 August 2023 to 13 June 2025; hard filtering by menu and allergens; offline evaluation with a temporal hold-out (training to 31 March 2025, testing 1 April to 13 June 2025); results reported overall and by behavioural segment. **Out of scope:** live deployment and A/B testing, nutrition optimisation, pricing, and kitchen forecasting. **Key variables** are the chosen entrée (target for RQ1 to RQ3), the cancellation flag (target for RQ4), and the predictors defined in the data dictionary and feature table. **Feasibility** is established in three ways: the data contain 2.42 million usable lunches, the required sample sizes are met by a wide margin (see Sample Size Determination), and the complete pipeline already runs on a laptop in under ten minutes.

Literature Survey

Literature Review Approach

Sources were identified in Google Scholar, the ACM Digital Library, and arXiv using the keywords *repeat consumption*, *next-basket recommendation*, *calibrated recommendation*, *cold-start*, *concept drift*, *food recommender*, *school meal waste*, *learning to rank*, and *synthetic data recommender*. Inclusion required peer-reviewed publication or a widely cited technical report or preprint, relevance to at least one research question or design decision, and a clear description of method and data. Sources are grouped by the research question they inform.

Summary of Key Literature

Table 2 summarises the twelve sources that most directly shaped the project; additional methodological sources are cited where they are used.

Table 2

Literature Relevance Matrix

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	How it supports RQs / decisions
Ren et al. (2019)	Session-based recommendation	E-commerce and music sessions	Encoder–decoder with explicit repeat-or-explore mode	Modelling repeat behaviour separately improves accuracy	RQ1: repetition must be modelled explicitly
Li et al. (2023)	Next-basket recommendation	Grocery purchase datasets	Benchmark separating repeat and explore items	Much reported gain comes from repeats; exploration is far harder	RQ1: repeat-favourite is the baseline to beat
Steck (2018)	Movie recommendation	Public movie-rating data	Re-ranking to match a user's historical genre mix	Calibration restores diversity at small accuracy cost	RQ1: basis for variety-calibrated re-ranking
Schein et al. (2002)	Cold-start recommendation	Public movie-rating data	Combining content and collaborative signals	Attribute information carries new users and items	RQ3: cohort and segment priors for new students
Gama et al. (2014)	Concept drift (survey)	Streaming learning tasks	Taxonomy of drift detection and adaptation	Models degrade when data relationships shift	RQ2: menu rotation treated as drift
Trattner & Elsweiler (2017)	Food recommendation (survey)	Recipe and meal platforms	Review of food recommender research	Field centres on adults; health and context remain open	Gap: no children's meal ordering
Byker Shanks et al. (2017)	School meal programmes	US school lunch studies, 1978–2015	Systematic review of plate waste	Plate waste in school meals is substantial	Motivation: wanted meals reduce waste
Gilmour & Fairchild (2024)	Public-health data access	UK school canteen transaction data	Case study of data acquisition, 2022–2024	Access blocked by ownership, agreements, and sensitivity	Data: justifies a calibrated synthetic dataset
Assylbekov et al. (2023)	Food delivery	About 4 million real orders, 3 cities	Public benchmark dataset	Repeat ordering is common in real food data	Planned external validation of RQ1
Stavinova et al. (2022)	Recommender evaluation (survey)	Synthetic-data simulators	Review of simulation-based evaluation	Simulation enables evaluation against known truth	Data: ground-truth recovery checks
Burges (2010)	Learning to rank	Web search ranking	RankNet, LambdaRank, LambdaMART	Gradients can directly optimise ranking metrics	Model 3: LambdaRank objective
Ke et al. (2017)	Gradient boosting	Public benchmark datasets	LightGBM: histogram-based boosted trees	Fast training with accuracy comparable to other boosting	Model 3: scalable, interpretable importance

Repeat behaviour and variety (RQ1). Two families of approach address repeat consumption. Neural models such as RepeatNet learn when a user will repeat and when they will explore (Ren et al., 2019), whereas simple frequency rules are strong baselines that complex models often fail to beat by much (Li et al., 2023). Both optimise accuracy, which tends to push recommendations toward a user's dominant interest; calibrated recommendation corrects this after ranking by matching the user's historical mix (Steck, 2018). The comparison suggests a hybrid design for this project: a learned ranker evaluated strictly against a repeat-favourite rule, followed by calibration, rather than a deep model whose gains over frequency rules are uncertain.

Changing menus and new students (RQ2 and RQ3). Drift research shows that models trained on past relationships degrade when those relationships shift, and recommends either detecting change or using features that generalise across it (Gama et al., 2014). Cold-start work shows that, until personal history accumulates, attribute or group-level information must carry predictions (Schein et al., 2002). Neither line of work has studied a catalogue that rotates on a fixed schedule, or how fast children's preferences become predictable.

Food, school meals, and data (motivation and data choice). Food recommender research focuses on adult recipe and restaurant choice and highlights health and context as open problems (Trattner & Elsweiler, 2017). School-meal research focuses on nutrition and waste rather than ordering (Byker Shanks et al., 2017). Real repeat-ordering data exist for adult food delivery (Assylbekov et al., 2023), but individual school transaction data are rarely accessible to researchers (Gilmour & Fairchild, 2024). Simulation-based evaluation is the accepted response, with the added benefit that results can be checked against known behaviour (Stavinova et al., 2022).

Methods for evaluation and segmentation. The studies above differ in how they evaluate. Most recommender studies score models offline on held-out interactions, and the next-basket literature shows that random splits overstate performance when behaviour is habitual, because the test set then contains the same routines as the training set (Li et al., 2023). A

temporal split, training on earlier months and testing on later ones, is the more realistic design and is adopted here. For describing users, centroid-based clustering remains the standard starting point because it is scalable and its segments are easy to explain (MacQueen, 1967), with internal validity judged by indices such as the silhouette (Rousseeuw, 1987). Hierarchical alternatives such as Ward's method optimise the same objective without random starts (Ward, 1963), which makes them a useful robustness check rather than a replacement.

Research gap. No study combines repeat consumption, a small scheduled-rotation menu, hard safety constraints, and children as consumers. It is unknown how much a learned ranker improves on a repeat rule in this setting, whether gains differ by type of child, how quickly new students become predictable, and whether prediction quality explains auto-order cancellations. LambdaRank-based gradient boosting (Burges, 2010; Ke et al., 2017) is well suited to testing these questions at scale.

Data Description

The study uses a calibrated synthetic dataset that mirrors the operational database of a US school-lunch subscription platform (Pathak, 2026). The table and column structure follows an approved ten-row sample and the provider's data dictionary, and behaviour is calibrated to aggregate statistics from the real platform, such as the share of students without allergen records, weekly ordering frequency, and weekday-anchored menu items. No real student record is included; the provider is not named. Synthetic data were necessary because individual-level school meal data are rarely shared (Gilmour & Fairchild, 2024).

Data Source(s) and Access

- Repository (code, notebooks, processed data): https://github.com/kkpathak91/school-meal-autoorder
- Raw data (14 comma-separated values [CSV] tables, zipped): https://github.com/kkpathak91/school-meal-autoorder/releases/tag/v1.0
- Citation: Pathak, K. K. (2026). *School meal subscription synthetic dataset* (Version 1.0) [Data set]. GitHub. https://github.com/kkpathak91/school-meal-autoorder

Dataset Overview

- Number of records (rows): 2,716,972 lunch orders in the main table, plus 3,036,697 side-item records, about 980,000 menu rows, and 30,000 students.
- Number of variables (columns): 15 in the lunch table and 15 in the student table; 14 tables in total.
- Time period: 14 August 2023 to 13 June 2025 (two school years and one summer).
- Unit of analysis: one student on one delivery day (a prediction event); one student for segmentation.
- Target variables: (a) the entrée the student kept, MENUITEMID of the live lunch (RQ1 to RQ3); (b) whether a legacy auto-order was cancelled, LUNCHCANCELED not empty (RQ4).

The data cover 30,000 students in 105 schools and 227 packing groups, which are the kitchen groupings that receive a common menu. Of the 2,716,972 lunch records, 2,420,160 are live and chosen by families, which forms the analysis base; the rest are cancelled orders and emergency lunches, which schools hand out on credit when no order exists and record as IOU ('I owe you') lunches.

The fourteen tables form a relational structure. The lunch table records every order and links to the student table through STUDENTID and to the grade table through GRADEID. Each grade belongs to a serving group, and each serving group to a packing group, which receives one published menu per delivery date in the menu table; this chain is what defines the set of entrées a student could have chosen on a given day. Item attributes come from the menu-item and cuisine tables, side items from the add-on and component tables, and allergen constraints from the student-allergen and item-allergen tables. A separate ground-truth folder records each simulated student's hidden traits; it is used only to check whether the segmentation recovers real behaviour, never as a model input.

Data Dictionary (Mandatory)

Table 3 defines the variables used in the analysis, with their data types, allowed values, and how missing values were handled; the full column list of every table is documented in the repository's data card.

Table 3

Data Dictionary

Variable name	Definition	Datatype	Allowed values / range	Missing values handling	Notes
STUDENTLUNCHID	Unique lunch order	Integer	70,000,001+	None missing	Primary key
STUDENTID	Student receiving the lunch	Integer	9,500,001–9,530,000	None missing	Grouping unit
GRADEID	Grade at the time of the lunch	Integer	Grade table keys	None missing	Links to school and packing group
MENUITEMID	Entrée ordered	Integer (categorical)	87 items	None missing	Target for RQ1–RQ3
PORTIONCODE	Portion size	Integer (categorical)	1 = Regular, 2 = Large	None missing	Large only where offered
DELIVERYDATE	Date the lunch is served	Date	2023-08-14 to 2025-06-13	None missing	Defines train/test split
STUDENTTYPE	Plan or retail type on that date	String (categorical)	Paid, Free, Staff, Sub DSU0	None missing	Plan code: days, sides, upgrade
LUNCHTYPE	How the lunch was created	String (categorical)	PreOrder, AutoOrder, LastMinute, IOU	None missing	IOU excluded from modelling
ORDERCHANNEL	Device or site used	String (categorical)	app iphone, app android, web calendar, web mobile	10.2% empty: system orders; kept as 'none'	Structural, not missing at random
LUNCHCREATED	Order creation time	Datetime	Before delivery	None missing	Lead-time feature
LUNCHCANCELED	Cancellation time	Datetime	Empty or timestamp	89.3% empty = live lunch	Target for RQ4
ISSERVED	Deprecated served flag	Boolean	Always false	Dropped	Carries no information
NSLPELIGIBILITY	Deprecated eligibility field	String	Always empty	Dropped (100% empty)	Export artifact

Variable name	Definition	Datatype	Allowed values / range	Missing values handling	Notes
CUISINEID	Cuisine of the entrée	Integer (categorical)	10 cuisines	None missing	Item attribute
ADVENTUROUSRATING	How unusual the entrée is	Integer (ordinal)	0–3	None missing	Item attribute
SERVINGTYPE	Hot or cold entrée	String (binary)	Hot, Cold	None missing	Item attribute
COMPONENTID (add-on)	Side item attached to a lunch	Integer (categorical)	35 sides	None missing	Side-item behaviour
PALETTERATING	Stored adventurousness of student	Integer (ordinal)	0–3	None missing	Round 1 clustering input
DEFAULTPORTIONCODE	Student's default portion	Integer (categorical)	0, 1, 2	None missing	Round 1 clustering input
DATEREGISTERED	Account registration time	Datetime	2015–2025	None missing	Identifies new students (RQ3)
ACTIVECODE	Account active	Boolean	true, false	None missing	Population filter
STUDENTTYPEEFFECTIVE	Date current type began	Date	Date or empty	36.6% empty; not used	Plan dates recoverable from lunches
ALLERGENID (student)	Active allergen or diet preference	Integer (categorical)	15 codes	No record = no allergen	Hard constraint filter
GRADE	Grade label	String (ordinal)	TK, K, 1–12, Staff	None missing	Mapped to age band
PACKINGGROUPID (menu)	Kitchen grouping with a common menu	Integer	227 groups	None missing	Defines candidate set

Sample Size Determination

Each research question needs enough independent students to detect a practically relevant effect. A power calculation is used where a question tests a difference (RQ1, RQ2, RQ4), and a confidence-interval (precision) calculation where it estimates a quantity (RQ3), with significance level $\alpha = .05$ (two-sided) and power $1 - \beta = .80$, so $z_{1-\alpha/2} = 1.960$ and $z_{1-\beta} = 0.8416$. Because each student contributes several events, event counts for RQ2 and RQ4 are inflated by the design

effect DEFF = 1 + (m − 1)ρ (Kish, 1965), where m is events per student and ρ the intra-class correlation, then converted to students.

RQ1, paired comparison, small effect d_z = 0.20 (Cohen, 1988), with Guenther's (1981) correction: $n = [(1.960 + 0.8416) / 0.20]^2 + 1.960^2/2 = 196.2 + 1.9 = 198.1 \rightarrow$ **199 students**.

RQ2, two proportions p_1 = .45 vs p_2 = .40 (Fleiss et al., 2003): $n = [1.960\sqrt{(2 \times .425 \times .575)} + 0.8416\sqrt{(.2475 + .2400)}]^2 / .05^2$ = 1,534 events per phase; DEFF = 1 + 4(.30) = 2.2 → 3,375 events ÷ 5 per student = **675 students**.

RQ3, 95% confidence interval of ±.03 for a proportion at the conservative p = .50: $n = 1.960^2 \times .25 / .03^2 = 1{,}067.1 \rightarrow$ **1,068 new students**.

RQ4, two proportions .20 vs .25: $n = [1.960\sqrt{(2 \times .225 \times .775)} + 0.8416\sqrt{(.16 + .1875)}]^2 / .05^2$ = 1,094 per group; DEFF = 1 + 9(.10) = 1.9 → 4,158 lunches ÷ 10 per student = **416 students**.

Table 4

Required and Available Sample Sizes

RQ	Method	Required (students)	Available in the data
RQ1	Power, paired comparison	199	2,273 sampled students with test-window lunches
RQ2	Power, two proportions with DEFF	675	16,880 students with test-window lunches
RQ3	Confidence interval for a proportion	1,068	5,874 new students with at least 10 orders
RQ4	Power, two proportions with DEFF	416	1,852 students with at least 10 auto-orders

Final N = max(199, 675, 1,068, 416) = 1,068 students, met for every research question

Table 4 compares each requirement with the data available. The largest requirement is chosen because one sample must answer all four questions. The RQ1 models were trained on a stratified sample of 3,000 students, of whom 2,273 had test-window lunches, which exceeds the RQ1 requirement eleven-fold.

GitHub Data Availability Statement

The raw and processed datasets are available in the GitHub repository under:

- Raw data: *data/raw/* (download *school_meal_synthetic_v2.zip* from release v1.0 and unzip here; the tables exceed GitHub's 100 MB file limit, so they are distributed as a release asset)
- Processed/clean data: *data/processed/student_features_clusters.csv* (21,668 students, 11 engineered features, and segment labels)
- Code/notebooks: *src/* (data generator and validation) and *notebooks/* (EDA, features, clustering, and models)

Screenshots / Evidence

Figure 1 previews the main lunch table and Figure 2 shows the repository structure.

Figure 1

Preview of the Raw Lunch Table

data/raw/STUDENTLUNCH.csv: first 8 of 2,716,972 rows (11 of 15 columns shown)

STUDENTLUNCHID	STUDENTID	GRADEID	MENUITEMID	PORTIONCODE	DELIVERYDATE	STUDENTTYPE	LUNCHTYPE	ORDERCHANNEL	LUNCHCREATED	LUNCHCANCELED
70000001	9503246	9300183	5010	1	2023-08-29	Paid	PreOrder	web mobile	2023-06-28 18:45:13	
70000002	9503246	9300183	5059	1	2023-08-30	Paid	PreOrder	web mobile	2023-06-28 18:45:13	
70000003	9503246	9300183	5058	1	2023-08-31	Paid	PreOrder	web mobile	2023-06-28 18:45:13	
70000004	9501661	9300102	5001	1	2023-08-23	Paid	PreOrder	app iphone	2023-06-28 19:01:32	
70000005	9529151	9302031	5023	1	2023-08-21	Sub 5010	PreOrder	app iphone	2023-06-28 19:18:13	
70000006	9529151	9302031	5011	1	2023-08-22	Sub 5010	PreOrder	web calendar	2023-06-28 19:18:13	
70000007	9529151	9302031	5018	2	2023-08-23	Sub 5010	PreOrder	app iphone	2023-06-28 19:18:13	
70000008	9529151	9302031	5075	1	2023-08-24	Sub 5010	PreOrder	app iphone	2023-06-28 19:18:13	

Figure 2

Repository Folder Structure

Repository structure: github.com/kkpathak91/school-meal-autoorder

```
school-meal-autoorder/
├── README.md                          project overview and run instructions
├── requirements.txt                   pinned Python packages
├── data/
│   ├── README.md                      how to obtain the raw data (Release v1.0)
│   ├── raw/                           14 CSV tables + _ground_truth/ (from Release zip)
│   └── processed/
│       └── student_features_clusters.csv   21,668 students x 11 features + cluster
├── src/
│   ├── generate_school_meal_data.py   reproducible data generator (seed 42)
│   └── validate_dataset.py            rule checks and calibration report
├── notebooks/
│   ├── 02_eda_by_hypothesis.ipynb     baseline EDA for H1-H4
│   └── 03_EDA_Features_Clustering.ipynb   cleaning, EDA, features, clusters, models
└── docs/
    ├── DATA_CARD.md                   dataset documentation
    └── VALIDATION_REPORT.txt          generator validation output
```

Analysis

Data Cleaning

Cleaning followed one principle: keep every record that reflects genuine behaviour, and remove only records that carry no information about a family's choice. Missing values, duplicates, and outliers were each measured before any decision; Table 5 logs every step with counts and percentages.

Table 5

Data Cleaning Log

Issue	Variables affected	Detection method	Treatment applied	Rationale
Missing values (structural)	LUNCHCANCELED	Empty-value count: 2,426,741 (89.3%)	Retained; empty = live lunch	Absence of a cancellation is the information itself
Missing values (structural)	ORDERCHANNEL	Export placeholder count: 277,086 (10.2%)	Recoded to 'none'	Auto-orders and emergency lunches have no ordering device
Missing values (empty fields)	NSLPELIGIBILITY, DATEUPDATED	100% empty	Dropped	No information

Issue	Variables affected	Detection method	Treatment applied	Rationale
Missing values (partial)	ORDERCHANNELDETAIL	Export placeholder count: 600,169 (22.1%)	Dropped	Ordering-screen detail; not needed for any RQ
Missing values (partial)	STUDENTTYPEEFFECTIVE	36.6% empty	Not used	Plan dates are recoverable from lunch-level plan codes
Constant column	ISSERVED	One value (false) in 100% of rows	Dropped	Deprecated field; no variance
Duplicates	All tables	Exact-row and key checks	None found (0 rows)	Data integrity confirmed
Duplicates (logical)	STUDENTID × DELIVERYDATE	More than one live lunch per student-day	None found (0)	One lunch per child per day
Outliers	Order lead time (hours)	Interquartile range (IQR) rule, 1.5 × IQR	1,688 (0.07%) retained	Valid monthly planners ordering four weeks ahead
Outliers	Lunches per delivery day	IQR rule	52 days (12.0%) retained	Holiday and summer-camp days; real seasonality
Outliers	Lunches per student	IQR rule	212 students (0.8%) retained	Heavy daily users are valid
Outliers	Engineered features	IQR rule	Retained; Yeo–Johnson transform for skewed features	Reduces influence without discarding behaviour
Non-choice records	LUNCHTYPE = IOU	Type filter: 6,581 rows	Excluded from modelling	Emergency lunches are chosen by staff, not families
Non-choice records	Cancelled lunches	290,231 rows (10.7%)	Excluded as choice labels; kept for RQ4	A cancelled order is not the lunch eaten
Sparse histories	Students	3,320 never ordered; 4,902 with fewer than 10 training orders	Excluded from segmentation; covered by RQ3	Proportions from fewer than 10 orders are unstable

Two decisions deserve explanation. First, outliers were retained rather than removed. Every extreme value checked corresponded to real behaviour, such as families ordering a month ahead or school holidays, and removing them would bias the data toward an average family that does not exist. Their influence was instead limited by rank-based statistics and by a Yeo–Johnson power transformation of skewed features (Yeo & Johnson, 2000). Second, segmentation is

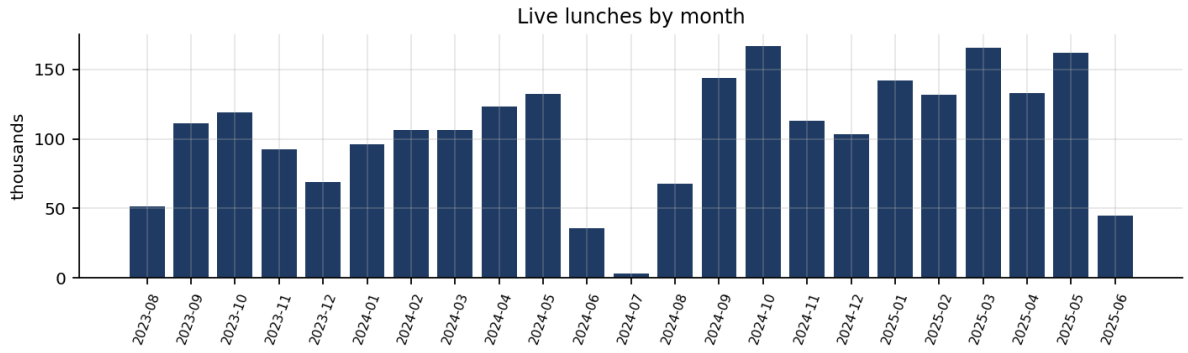
limited to students with at least ten training orders, because the standard error of a proportion, $\sqrt{[p(1-p)/n]}$, reaches 0.16 at $n = 10$ and 0.22 at $n = 5$. The workflow is fully reproducible: every step runs from the notebooks with a fixed random seed (Appendix C).

EDA Results (Interpretation Emphasis)

Volume and seasonality. As shown in Figure 3, lunch volume follows the school calendar: monthly totals collapse during summer, when only camps operate, and dip at Thanksgiving, winter break, and spring break. A typical school day carries about 6,021 lunches. Volume rises in 2024–25 as new students enrol. The insight is that time periods are not interchangeable, which is why the evaluation uses a temporal split rather than a random one.

Figure 3

Live Lunches by Month



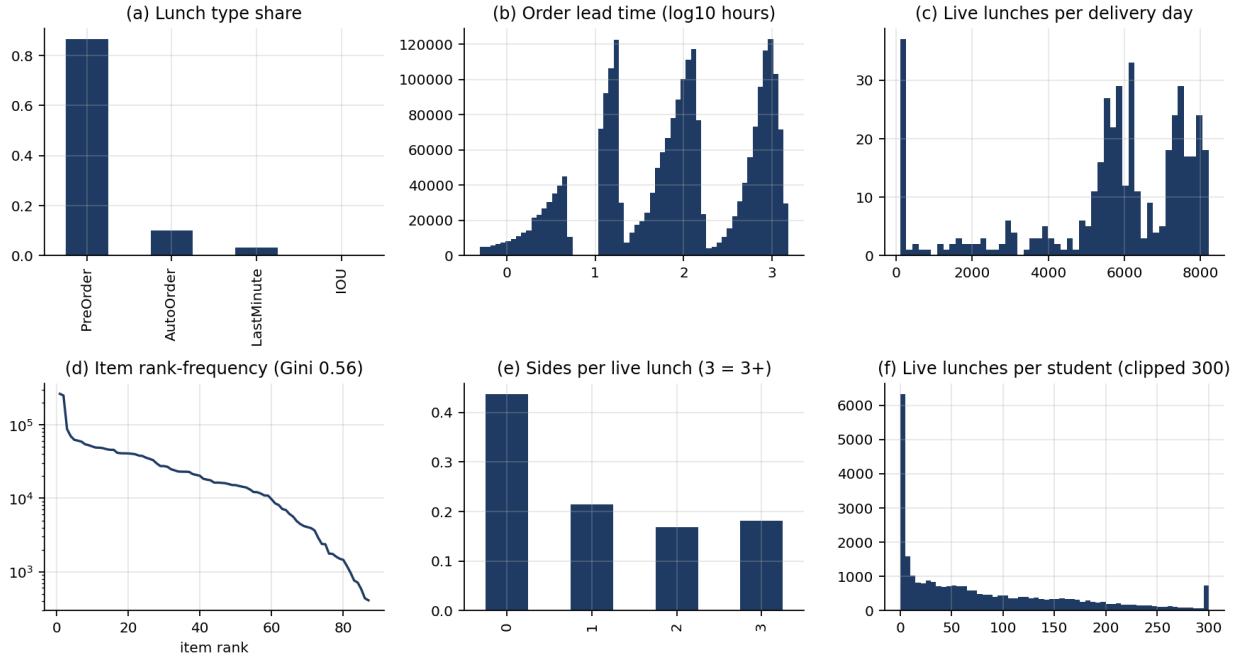
Distributions of lunch-level variables. Figure 4 shows that demand is concentrated: the Gini coefficient of item popularity is 0.56, and the ten most popular entrées take 42.1% of lunches (panel d). Order lead time is right-skewed (median 94 hours, skew 1.18), and 43.6% of lunches carry no side item (panel e). Activity per student is highly unequal (Gini 0.56), and 11.1% of registered students never ordered. Concentration on a few items is the first sign that a repeat-favourite rule will be a strong baseline for RQ1.

Two further patterns shape the design. Weekday volume is not uniform ($\chi^2(4) = 7{,}741$, $p < .001$): Mondays and Fridays carry fewer lunches because holidays and some school closures fall on those days, so weekday is included as a feature and weekday habits are measured per student.

Side items are strongly tied to subscriptions, with plan lunches carrying 1.64 sides on average against 0.55 for retail lunches, which means side behaviour separates subscribers from other families and belongs in the segmentation rather than in entrée prediction.

Figure 4

Univariate Distributions of Lunch-Level Variables

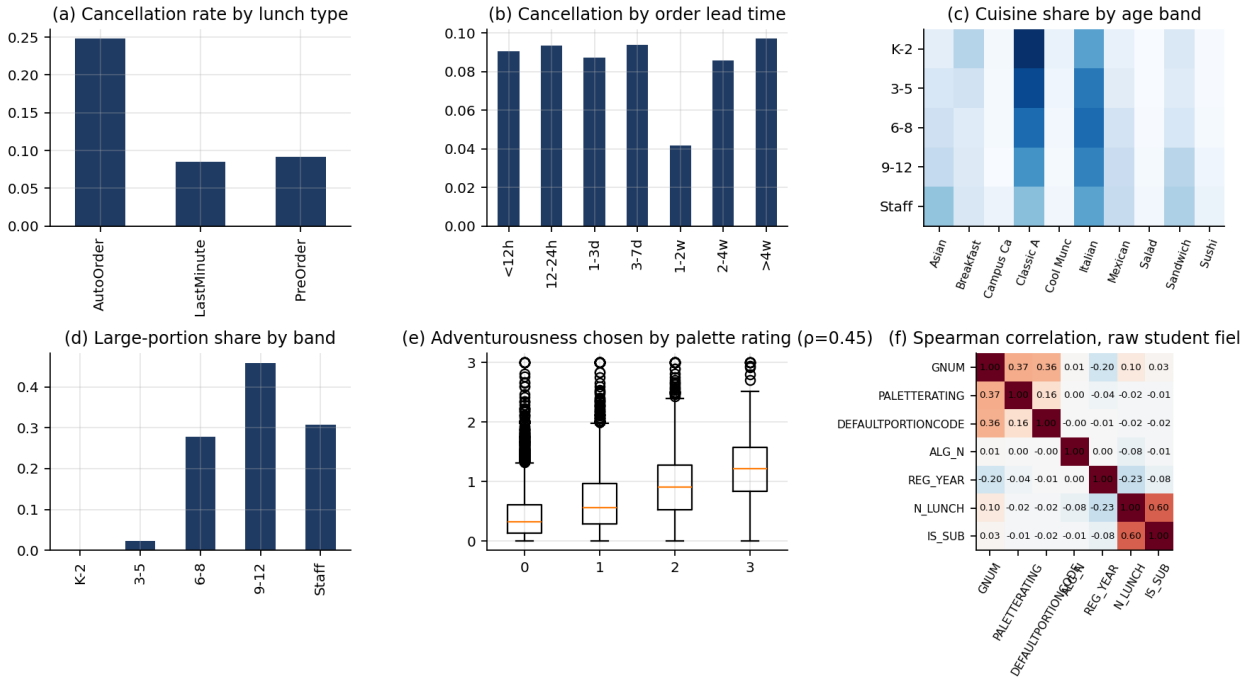


Note. (a) lunch types; (b) lead time, log10 hours; (c) lunches per day; (d) item rank-frequency, log scale; (e) sides per lunch; (f) lunches per student.

Relationships between raw variables. Figure 5 and Table 6 test pairs of variables with methods matched to their types (formulas in Appendix B). With 2.4 million records almost every test is significant, so interpretation relies on effect sizes. Three relationships matter most. Auto-orders are cancelled almost three times as often as pre-orders (Cramér's V = 0.15), which motivates RQ4. Age shapes choice: cuisine mix shifts with age band (V = 0.13) and older students choose more adventurous food (Spearman ρ = 0.65). Order lead time, by contrast, has no meaningful link with cancellation (rank-biserial r = -0.007), so it was excluded as a feature.

Figure 5

Bivariate Relationships in the Raw Data



Note. (a) cancellation by lunch type; (b) by lead time; (c) cuisine by age band; (d) large portions by age band; (e) chosen adventurousness by stored palette rating; (f) Spearman correlations of stored student fields.

Table 6

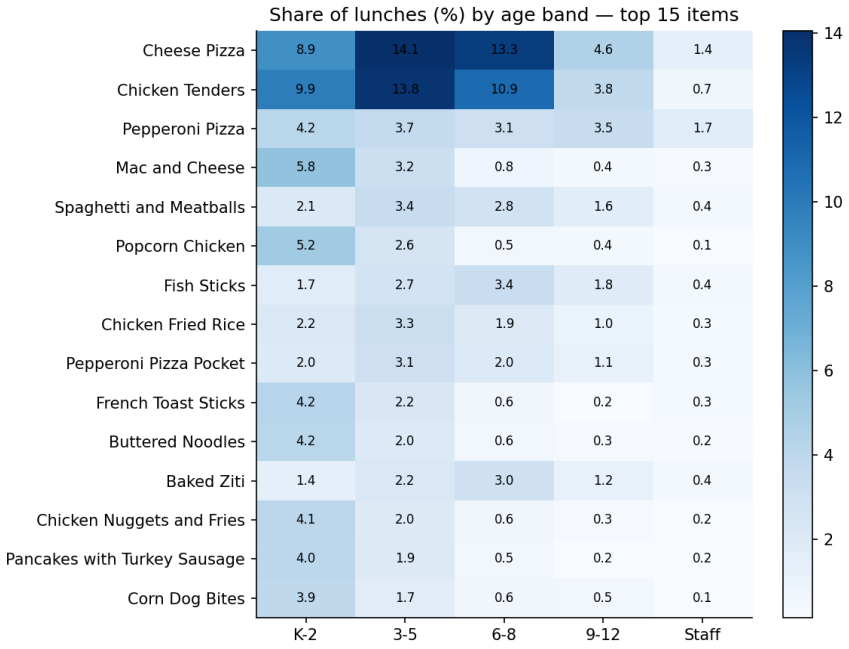
Bivariate Tests on Raw Variables

Pair	Test	Statistic	Effect size	Interpretation
Lunch type × cancellation	Chi-square	$\chi^2(2) = 62{,}288$	V = 0.15	Auto-orders 24.8% vs pre-orders 9.2%
Lead time × cancellation	Mann–Whitney U	p = 0.05	r = -0.007	Negligible; feature dropped
Age band × cuisine	Chi-square	$\chi^2(36) = 164{,}118$	V = 0.13	Preferences change with age
Age band × portion	Chi-square	$\chi^2 = 554{,}284$	V = 0.48	Structural: Large not offered to young grades
Subscription × sides	Mann–Whitney U	p < .001	r = -0.52	Plans include sides (1.64 vs 0.55)
Grade × adventurousness	Spearman	p < .001	ρ = 0.65	Older students explore more

Age-related preferences. Figure 6 shows that the youngest students concentrate on a few familiar items such as pizza, tenders, and macaroni, whereas older students spread across bowls, sandwiches, and sushi. Age band is therefore a necessary feature and a necessary reporting dimension for every research question.

Figure 6

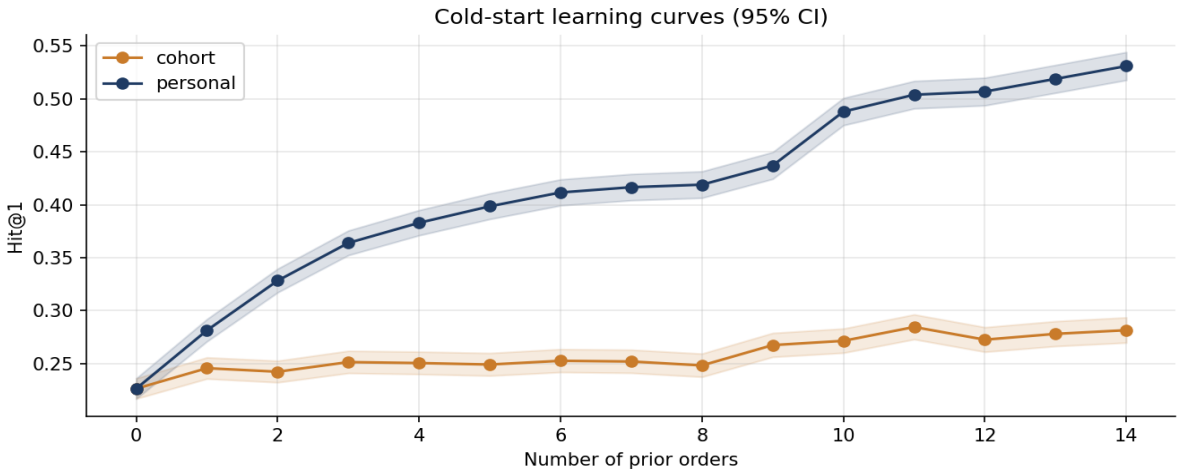
Share of Lunches by Age Band for the 15 Most-Ordered Items



Hypothesis-driven baselines. Transparent rules were scored on the test window (1 April to 13 June 2025) to check that each research question has a measurable effect. For RQ1, repeating each student's most frequent eligible item reaches Hit@1 = 0.588 against 0.348 for school-level popularity and 0.143 for random choice. For RQ2, repeat-favourite accuracy falls from 0.590 mid-cycle to 0.573 in the ten school days after a menu change ($z = 7.25$, $p < .001$). For RQ3, Figure 7 shows that among 8,286 students who joined in 2024–25, a student's own history beats the cohort guess from the first prior order and keeps improving, reaching 0.531 by the fourteenth. For RQ4, legacy auto-orders whose pick differs from the student's usual choice are cancelled at 29.0% against 20.9% for matching picks (odds ratio 1.55).

Figure 7

Cold-Start Learning Curves: Personal History Versus Cohort Guess

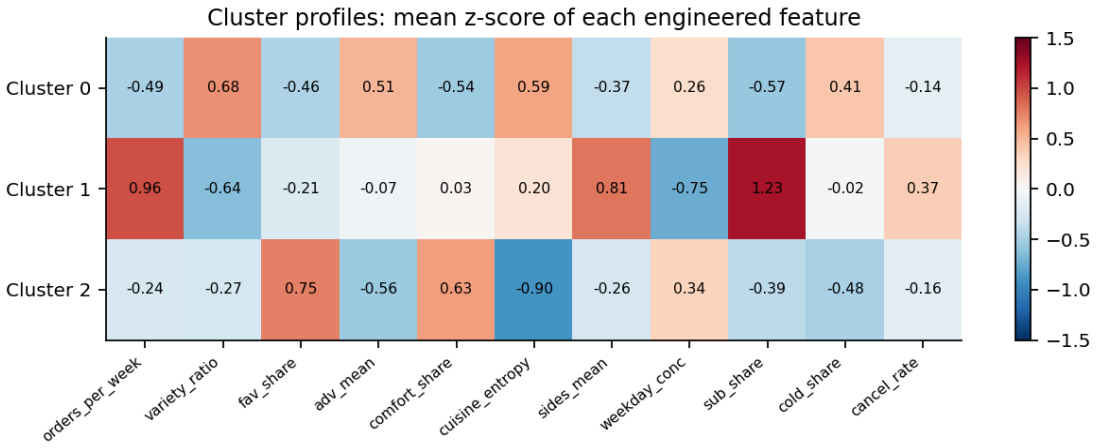


Note. Shaded bands are 95% Wilson confidence intervals.

Behavioural segments. K-means segmentation on engineered features (Modelling, Model 1) produced three segments whose profiles are shown in Figure 8. *Adventurous explorers* (39% of students) have the highest variety and adventurousness and rarely subscribe; *committed subscribers* (28%) order 2.7 times a week on plans with many sides; *comfort-food loyalists* (32%) place 45% of orders on one favourite and 87% on comfort cuisines, and three quarters are in kindergarten to grade 5. Figure 9 shows that the segments differ on every research question, which is the most important EDA insight: one rule cannot serve all children equally well.

Figure 8

Segment Profiles as Mean z-Scores of the Engineered Features



Note. Cluster 0 = adventurous explorers; 1 = committed subscribers; 2 = comfort-food loyalists.

Figure 9

Research-Question Outcomes by Segment

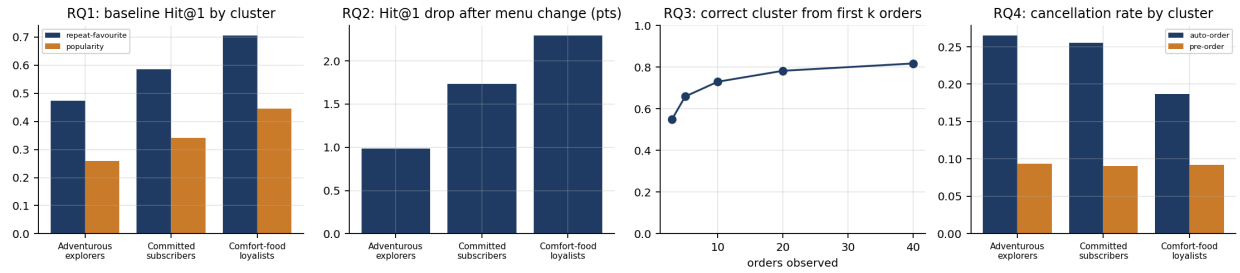


Table 7 summarises the insight carried by each exhibit and the decision it drove.

Table 7

EDA Insight Summary

Figure/Table reference	What it shows	Key insight	Why it matters (link to RQ/objective)	Decision / next step
Figure 3	Monthly volume	Strong school-calendar seasonality	Periods are not exchangeable	Temporal train/test split
Figure 4	Univariate distributions	Few items dominate; activity is unequal	Repeat rule is a strong baseline (RQ1)	Repeat-favourite as primary comparator
Figure 5, Table 6	Bivariate tests	Auto-orders cancel most; lead time irrelevant	Defines RQ4; prunes features	Drop lead time; model cancellation
Figure 6	Items by age band	Preferences change with age	All RQs vary by age	Age band as feature and lens
Figure 7	Cold-start curves	Personal history helps from order 1	RQ3 threshold is low	Model-based cold start next
Figures 8–9	Segments and their outcomes	Repeat works for loyalists, fails explorers	RQ1, RQ2, RQ4 differ by segment	Report and calibrate by segment

Modelling

Choice of Models with Justification

The problem combines three tasks: ranking a small set of candidates (RQ1 to RQ3), binary classification (RQ4), and unsupervised segmentation. Models were chosen to match each task, to be explainable to a caterer, and to be compared against transparent baselines. **Baselines** are popularity (most-ordered item at the student's school and age band), repeat-favourite (the student's most-ordered eligible item), and, for RQ4, the overall cancellation rate and a simple mismatch rule.

Model 1: K-means segmentation. K-means partitions students to minimise within-cluster squared distance (MacQueen, 1967). It suits continuous, standardised behavioural features, scales to 21,668 students, yields readable centroids, and lets new students be assigned to the nearest centroid. Its assumptions are roughly spherical clusters of similar spread, so a Gaussian mixture model and Ward's hierarchical clustering (Ward, 1963) were run as checks. Segments were used as an analysis lens and tested as model features.

Model 2: Logistic regression. For RQ1 it scores each candidate item independently (pointwise), and for RQ4 it estimates the probability of cancellation. It is fast, its coefficients are directly interpretable, and it is the standard reference model. Its assumptions are a linear relationship between features and log-odds and independent observations; it cannot capture interactions unless they are specified.

Model 3: Light Gradient Boosting Machine (LightGBM). For RQ1 to RQ3, a LightGBM ranker with the LambdaRank objective learns to order each day's candidates by directly optimising a ranking metric (Burges, 2010; Ke et al., 2017); for RQ4, a LightGBM classifier predicts cancellation. Gradient-boosted trees capture non-linear effects and interactions, handle mixed feature types without scaling, and provide feature importance. Their limitation is lower transparency than logistic regression and a risk of overfitting, controlled here by early stopping on a time-ordered validation set. Neural sequence models such as RepeatNet (Ren et al., 2019) were considered but deferred, because the literature shows uncertain gains over frequency rules (Li et al., 2023) and the tree models already provide a strong, explainable benchmark.

Features Included and Feature Engineering

Fifteen behavioural features were engineered for segmentation from each student's training-period orders, then screened in three steps (Appendix A gives the full results). Pairs with Spearman $|\rho| > 0.75$ (more than 56% shared rank variance) were reduced to one member, removing repeat rate ($\rho = 0.81$ with favourite share) and item entropy ($\rho = -0.78$). Two features were removed on evidence from the EDA: lead time (no link with cancellation) and large-portion share (determined by age, $V = 0.48$). The remaining eleven have variance inflation factors (VIF) of 1.06 to 2.48, well below the threshold of 5 (O'Brien, 2007). Skewed features were Yeo–Johnson transformed and all were standardised. Candidate-level features for the ranking models describe how a student relates to each item on the menu that day. Table 8 lists the main features and where they are used.

Table 8

Feature Set and Usage

Feature	Original / Engineered	Type	Reason for inclusion	Used in model(s)
fav_share: max item share	Engineered	Continuous [0, 1]	Loyalty to one item	M1; M2/M3 (RQ1)
variety_ratio: distinct items ÷ orders	Engineered	Continuous [0, 1]	Breadth of choice	M1; M2/M3 (RQ1)
orders_per_week	Engineered	Continuous	Engagement	M1
adv_mean: mean adventurous rating	Engineered	Continuous [0, 3]	Willingness to explore	M1; M2/M3
comfort_share, cuisine_entropy	Engineered	Continuous	Cuisine preference and spread	M1
sides_mean, sub_share, weekday_conc, cold_share, cancel_rate	Engineered	Continuous	Plan, timing, and change behaviour	M1
hist_share: student's share of the candidate item	Engineered	Continuous	Personal preference	M2, M3 (RQ1)
dow_share: same, on that weekday	Engineered	Continuous	Weekday habits	M2, M3 (RQ1)
days_since: days since last chosen	Engineered	Count (0–400)	Recency and fatigue	M2, M3 (RQ1)
pop_share: item share at school and age band	Engineered	Continuous	Peer popularity	M2, M3 (RQ1, RQ4)
adv_gap: |item − student adventurousness|	Engineered	Continuous	Fit to the student's taste	M2, M3 (RQ1)
ADVENTUROUSRATING, SERVINGTYPE, cuisine group	Original	Ordinal, binary	Item attributes	M2, M3
Segment label (one-hot)	Engineered	Categorical	Behavioural type	M2, M3 (RQ1, RQ4)
match, pick_share, pref_share, share_gap	Engineered	Binary, continuous	Fit of the auto-order pick	M2, M3 (RQ4)
plan days, plan sides (from STUDENTTYPE)	Original (decoded)	Ordinal	Plan structure	M2, M3 (RQ4)
weekday, month	Original	Categorical (one-hot)	Calendar effects	M2, M3 (RQ4)

Note. M1 = k-means, M2 = logistic regression, M3 = LightGBM.

Leakage control. Training events (January to March 2025) use features computed only from history up to 31 December 2024, and test events (April to June 2025) use history up to 31 March 2025. Segments are fitted on training data only. Students with no earlier history receive zero history shares and training-median traits, so cold-start cases are kept rather than dropped. **Feature importance** (Figures 10 and 11) confirms the design: the student's history share of an item accounts for 66% of the ranker's gain and weekday habit for 15%, while segment labels add almost nothing once student-level features are present.

Evaluation Metrics (Include Formulae and Calculations)

Validation uses a temporal hold-out that mirrors deployment: models learn from the past and are scored on the following months. Within training, the last 15% of events by date served as an early-stopping validation set. Table 9 defines each metric; worked calculations use the preliminary results.

Table 9

Evaluation Metrics and Formulae

Metric	Formula	Interpretation	Used for
Hit@1	$(1/\|E\|)\ \Sigma_e$ 1[top-ranked item = chosen item]	Share of days the auto-ordered lunch is right; 0–1	RQ1–RQ3 (primary)
NDCG@3	$\Sigma_{r\le 3}\ \text{rel}_r / \log_2(r+1) \div$ ideal value	Credit for the chosen item in the top three; 0–1	RQ1, RQ2
Accuracy	$(TP + TN) / N$	Share of correct cancel / keep calls	RQ4
Precision	$TP / (TP + FP)$	Share of flagged orders that are cancelled	RQ4
Recall	$TP / (TP + FN)$	Share of cancellations that are flagged	RQ4
F1-score	2 × Precision × Recall / (Precision + Recall)	Balance of precision and recall	RQ4
AUC	P(score of a cancelled order > score of a kept order)	Threshold-free ranking quality; 0.5 = chance	RQ4 (primary)
PR-AUC	Area under the precision–recall curve	Minority-class quality; baseline = cancellation rate	RQ4
Brier score	$(1/N)\ \Sigma\ (p_i - y_i)^2$	Probability accuracy; 0 = perfect	RQ4
Silhouette, Davies–Bouldin, Calinski–Harabasz, ARI	Appendix A	Segment quality and stability	Model 1

Note. TP, FP, TN, FN = true/false positives/negatives; NDCG = normalised discounted cumulative gain; AUC = area under the receiver operating characteristic curve; PR-AUC = precision–recall AUC; ARI = adjusted Rand index.

Worked calculations. The LightGBM ranker placed the chosen entrée first in 27,993 of 45,742 test events, so Hit@1 = 27,993 / 45,742 = 0.612. If the chosen item is ranked second, its NDCG@3 contribution is $1 / \log_2(3) = 0.631$, and zero if it is ranked fourth or lower. For RQ4, the LightGBM classifier, at a threshold of 0.23 chosen on training data, flagged 31,319 of 45,220 test auto-orders, of which 8,345 were cancelled (TP), with 22,974 false positives, 2,245 missed cancellations (FN), and 11,656 correct keeps (TN). Hence precision = 8,345 / 31,319 = 0.266, recall = 8,345 / 10,590 = 0.788, F1 = 2 × 0.266 × 0.788 / (0.266 + 0.788) = 0.398, and accuracy = (8,345 + 11,656) / 45,220 = 0.442.

Preliminary Results

Preliminary Findings by Research Question

RQ1 (completed: baselines and first models; remaining: variety calibration). Table 10 and Figure 10 show that the LightGBM ranker reaches Hit@1 = 0.612 against 0.590 for repeat-favourite and 0.345 for popularity, on 45,742 test events. The gain over repeat-favourite is significant: McNemar's test on events where only one method is right gives $\chi^2(1) = 213.2$, $p < .001$ (2,861 versus 1,857 discordant events), and a paired t test across 2,273 students gives $t = 8.18$, $p < .001$, $d_z = 0.17$. The null hypothesis is rejected, but the effect is small, which is consistent with the literature on strong frequency baselines. Gains are largest for explorers (+2.1 points) and subscribers (+3.0) and smallest for loyalists (+0.8).

Table 10

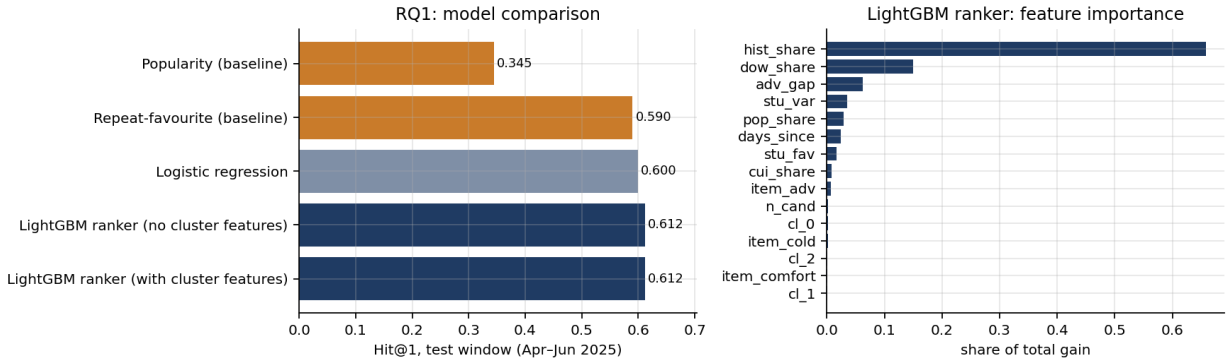
Preliminary Model Performance Comparison

Model	Feature set	Validation approach	Metric 1	Metric 2	Key takeaway
Popularity (baseline)	School and age-band popularity	Temporal hold-out, Apr–Jun 2025	Hit@1 0.345	NDCG@3 0.560	Weak: ignores the individual
Repeat-favourite (baseline)	Student history	Same	Hit@1 0.590	NDCG@3 0.732	Strong rule to beat
Logistic regression (RQ1)	15 candidate, student, and segment features	Same	Hit@1 0.600	NDCG@3 0.740	+1.0 point over rule
LightGBM ranker (RQ1)	12 features, no segments	Same, early stopping	Hit@1 0.612	NDCG@3 0.751	Best; +2.2 points
LightGBM ranker (RQ1)	15 features with segments	Same, early stopping	Hit@1 0.612	NDCG@3 0.750	Segments add no accuracy
Base rate (RQ4 baseline)	None	Temporal hold-out	AUC 0.500	PR-AUC 0.234	Reference
Logistic regression (RQ4)	20 auto-order features	Same	AUC 0.580	PR-AUC 0.286	Modest signal
LightGBM classifier (RQ4)	Same features	Same, early stopping	AUC 0.600	PR-AUC 0.299	Best; F1 0.398

The feature importance in Figure 10 explains why the gain is modest. Two thirds of the ranker's gain comes from the student's history share of each item, the same information the repeat-favourite rule uses, and a further 15% from weekday habit. The model's advantage lies in how it combines these with the fit between an item's adventurousness and the student's taste, recency, and peer popularity, which matters most when a favourite is missing or a student rotates between several items. This is why the gains concentrate among explorers and subscribers, whose choices a single favourite cannot describe. The segment labels add no accuracy once these student-level features are present (Hit@1 0.6120 with segments against 0.6122 without), so segments are more useful for reporting results and setting a variety target per group than as model inputs.

Figure 10

RQ1 Model Comparison and Ranker Feature Importance



RQ2 (completed: change detection and phase comparison; remaining: mixed-effects model and substitution features). Menu-cycle changes were detected from the menus themselves, as weeks in which a school's offerings differed from the previous two weeks by more than the 90th percentile of change; the detected weeks match the October, December, February, and April cycle starts. In the test window, every method loses accuracy after a change: LightGBM falls from 0.614 to 0.598 (1.7 points, $p = 0.011$) and repeat-favourite by 1.3 points. Across the full year, the drop grows with loyalty: 1.0 point for explorers, 1.7 for subscribers, and 2.3 for loyalists (Figure 9). H_0 is rejected; the model stays ahead after changes but does not yet

close the gap.

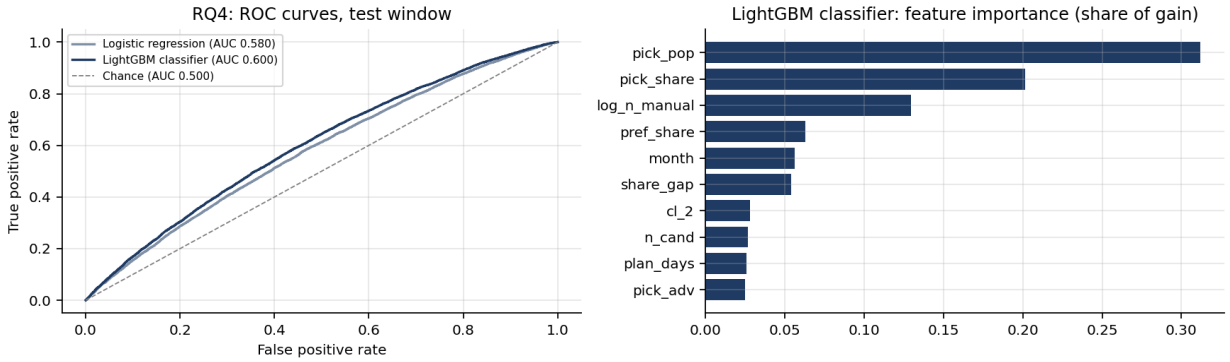
RQ3 (completed: baseline learning curves and segment assignment; remaining: model-based comparison). Personal history outperforms the cohort guess from the first prior order (Figure 7), so H_0 is rejected with $k^* = 1$. Assigning students to segments from their first orders recovers the full-history segment 55% of the time after 3 orders, 73% after 10, and 82% after 40, against 39% by chance, so segment priors can help during the first weeks.

RQ4 (completed: match analysis and two classifiers; remaining: calibration and cost analysis). Mismatched picks are cancelled more often (29.0% versus 20.9%), so H_0 is rejected. Predicting cancellation is harder: on 45,220 test auto-orders the LightGBM classifier reaches AUC = 0.600 against 0.580 for logistic regression (95% bootstrap interval for the difference, 0.014 to 0.025), and PR-AUC = 0.299 against a base rate of 0.234 (Figure 11). The most important features are the pick's popularity and the student's history with the picked item, so cancellations are driven mainly by whether the pick suits the child, the very quantity the RQ1 ranker improves.

Two cautions apply to the RQ4 results. The decision threshold that maximised F1 on training data is low (0.23), so the classifier flags most auto-orders and achieves high recall at low precision; in practice the threshold should reflect the relative cost of a missed cancellation and an unnecessary reminder to parents. The rule-based mismatch flag is a useful reference but produces only two score levels, so its probability-based metrics are not meaningful; its AUC of 0.549 nonetheless confirms that the match signal alone carries information.

Figure 11

RQ4 ROC Curves and Classifier Feature Importance



Note. ROC = receiver operating characteristic.

Segmentation Results

K-means with $k = 3$ was selected because it had the best Davies–Bouldin and Calinski–Harabasz indices and a stability adjusted Rand index of 0.99 across subsamples, and the Gaussian-mixture information criterion flattened after three components (Appendix A). Clustering on stored attributes alone produced segments defined by student type and grade; clustering on engineered features explained 2.7 times more of the variance in the hidden loyalty trait ($\eta^2 = 0.080$ versus 0.030), confirming that feature engineering captured behaviour rather than identity. Separation is moderate (silhouette 0.19), so the segments are best read as regions of a continuum.

Interim Limitations and Risks

- **Synthetic data (validity).** Results support the methods rather than claims about real children; models resembling the generator could score higher than on real data. *Mitigation:* repeat key results on a second generator seed and on the public food-delivery dataset (Assylbekov et al., 2023).
- **Small effect for RQ1 (reliability).** The ranker's gain is 2.2 points; tuning may change it. *Mitigation:* report confidence intervals and per-segment results, not only the overall figure.

- **Weak cancellation signal for RQ4.** AUC of 0.60 limits practical use. *Mitigation:* add plan-level and menu-change features and evaluate a cost-based threshold.
- **Dependence between events.** Lunches from one student are correlated. *Mitigation:* student-level paired tests, clustered bootstrap, and a planned mixed-effects model.
- **Segment separation.** Segments summarise a behavioural continuum (silhouette 0.19), so students near a boundary could reasonably belong to a neighbouring segment. *Mitigation:* use segments for reporting and group-level targets rather than hard rules, and report their stability.
- **Dietary preferences.** Beef, pork, vegetarian, and vegan preferences are not stored per menu item, so only true allergens are filtered. *Mitigation:* add a protein attribute before the final evaluation.

Next Steps (for Final Report)

- Add per-student and per-segment variety calibration to the RQ1 ranker (Steck, 2018), tune hyperparameters, and compare all models with a Friedman test.
- Add item-substitution features and fit a mixed-effects logistic model with phase × model and phase × segment terms for RQ2.
- Compare cohort, segment-prior, personal, blended, and LightGBM models on learning curves for RQ3.
- Improve and calibrate the RQ4 classifier, compare models with the DeLong test, and estimate cancellations under the proposed ranker.
- Run robustness checks on a second data seed and on the public food-delivery dataset.
- Write the final report with recommendations for the caterer, families, and schools.

Bibliography

Assylbekov, Y., Bali, R., Bovard, L., & Klaue, C. (2023). Delivery Hero Recommendation Dataset: A novel dataset for benchmarking recommendation algorithms. In *Proceedings of the 17th ACM Conference on Recommender Systems* (pp. 1042–1044). ACM.

Burges, C. J. C. (2010). *From RankNet to LambdaRank to LambdaMART: An overview* (Technical Report MSR-TR-2010-82). Microsoft Research.

Byker Shanks, C., Banna, J., & Serrano, E. L. (2017). Food waste in the National School Lunch Program 1978–2015: A systematic review. *Journal of the Academy of Nutrition and Dietetics, 117*(11), 1792–1807. https://doi.org/10.1016/j.jand.2017.06.008

Caliński, T., & Harabasz, J. (1974). A dendrite method for cluster analysis. *Communications in Statistics, 3*(1), 1–27. https://doi.org/10.1080/03610927408827101

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum Associates.

Davies, D. L., & Bouldin, D. W. (1979). A cluster separation measure. *IEEE Transactions on Pattern Analysis and Machine Intelligence, PAMI-1*(2), 224–227. https://doi.org/10.1109/TPAMI.1979.4766909

Fleiss, J. L., Levin, B., & Paik, M. C. (2003). *Statistical methods for rates and proportions* (3rd ed.). Wiley.

Gama, J., Žliobaitė, I., Bifet, A., Pechenizkiy, M., & Bouchachia, A. (2014). A survey on concept drift adaptation. *ACM Computing Surveys, 46*(4), Article 44. https://doi.org/10.1145/2523813

Gilmour, A., & Fairchild, R. (2024). The practical and ethical challenges of identifying, accessing and obtaining school canteen transactional data for public health research. *Digital Health, 10*. https://doi.org/10.1177/20552076241297356

Guenther, W. C. (1981). Sample size formulas for normal theory t tests. *The American Statistician, 35*(4), 243–244. https://doi.org/10.1080/00031305.1981.10479363

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems 30* (pp. 3146–3154).

Kish, L. (1965). *Survey sampling*. Wiley.

Li, M., Jullien, S., Ariannezhad, M., & de Rijke, M. (2023). A next basket recommendation reality check. *ACM Transactions on Information Systems, 41*(4), Article 116. https://doi.org/10.1145/3587153

MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability* (Vol. 1, pp. 281–297). University of California Press.

O'Brien, R. M. (2007). A caution regarding rules of thumb for variance inflation factors. *Quality & Quantity, 41*(5), 673–690. https://doi.org/10.1007/s11135-006-9018-6

Pathak, K. K. (2026). *School meal subscription synthetic dataset* (Version 1.0) [Data set]. GitHub. https://github.com/kkpathak91/school-meal-autoorder

Ren, P., Chen, Z., Li, J., Ren, Z., Ma, J., & de Rijke, M. (2019). RepeatNet: A repeat aware neural recommendation machine for session-based recommendation. *Proceedings of the AAAI Conference on Artificial Intelligence, 33*(1), 4806–4813. https://doi.org/10.1609/aaai.v33i01.33014806

Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics, 20*, 53–65. https://doi.org/10.1016/0377-0427(87)90125-7

Schein, A. I., Popescul, A., Ungar, L. H., & Pennock, D. M. (2002). Methods and metrics for cold-start recommendations. In *Proceedings of the 25th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval* (pp. 253–260). https://doi.org/10.1145/564376.564421

Stavinova, E., Grigorev, A., Volodkevich, A., Chunaev, P., Bochenina, K., & Bugaychenko, D. (2022). *Synthetic data-based simulators for recommender systems: A survey*

(arXiv:2206.11338). arXiv. https://doi.org/10.48550/arXiv.2206.11338

Steck, H. (2018). Calibrated recommendations. In *Proceedings of the 12th ACM Conference on Recommender Systems* (pp. 154–162). https://doi.org/10.1145/3240323.3240372

Trattner, C., & Elsweiler, D. (2017). *Food recommender systems: Important contributions, challenges and future research directions* (arXiv:1711.02760). arXiv. https://doi.org/10.48550/arXiv.1711.02760

Ward, J. H., Jr. (1963). Hierarchical grouping to optimize an objective function. *Journal of the American Statistical Association, 58*(301), 236–244. https://doi.org/10.1080/01621459.1963.10500845

Yeo, I.-K., & Johnson, R. A. (2000). A new family of power transformations to improve normality or symmetry. *Biometrika*, *87*(4), 954–959. https://doi.org/10.1093/biomet/87.4.954

Appendix A: Feature Screening and Segment Selection

Figure A1 shows the Spearman correlations used to remove redundant features; Table A1 the criteria used to choose the number of segments. Silhouette is $s(i) = [b(i) - a(i)] / \max\{a(i), b(i)\}$ (Rousseeuw, 1987); Davies–Bouldin is the mean over clusters of the worst ratio of within-cluster spread to between-centroid distance (Davies & Bouldin, 1979); Calinski–Harabasz is the ratio of between- to within-cluster variance adjusted for degrees of freedom (Caliński & Harabasz, 1974); stability is the mean adjusted Rand index between the full fit and fits on five 80% subsamples.

Figure A1

Spearman Correlation of the Fifteen Candidate Features

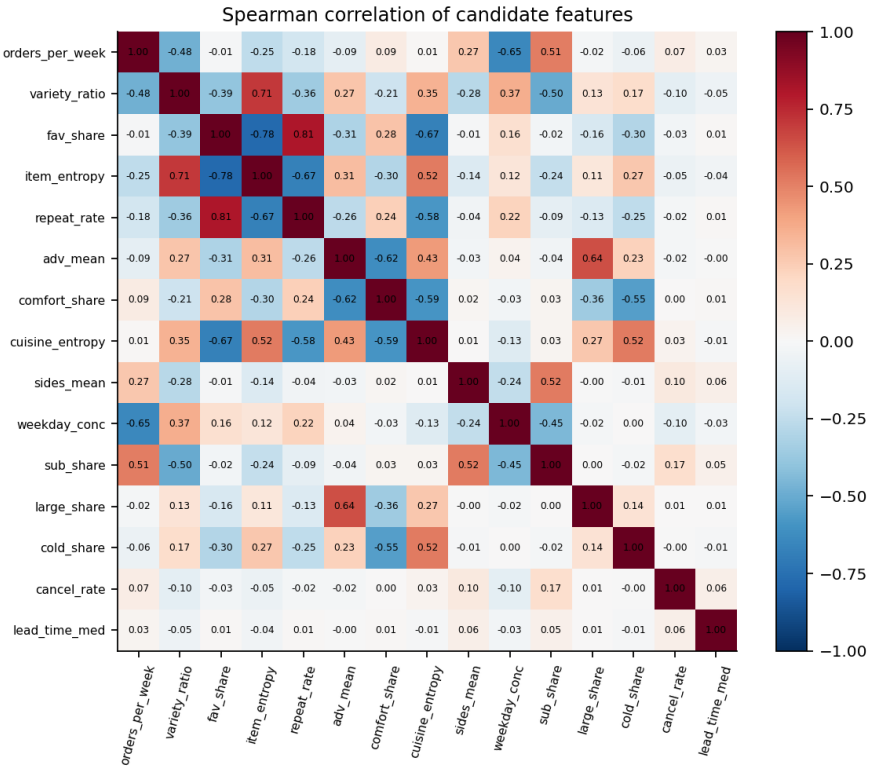


Table A1

Segment-Selection Criteria by Number of Clusters (Engineered Features)

K	Silhouette	Davies–Bouldin	Calinski–Harabasz	Stability ARI	Smallest cluster
2	0.191	1.940	5,031	0.825	37.6%
3	0.188	1.655	5,143	0.991	28.3%
4	0.155	1.839	4,145	0.980	16.8%

5	0.155	1.869	3,642	0.704	8.9%
6	0.125	1.935	3,311	0.974	7.8%
7	0.128	1.831	3,023	0.968	4.9%
8	0.128	1.834	2,807	0.907	4.8%
9	0.129	1.791	2,615	0.956	4.5%
10	0.130	1.753	2,450	0.770	4.2%

Appendix B: Statistical Tests Used

- **Chi-square test and Cramér's V:** $\chi^2 = \Sigma\,(O - E)^2/E$; $V = \sqrt{[\chi^2/(N(\min(r, c) - 1))]}$. Used for categorical pairs; effect sizes guide interpretation because of the very large sample.
- **Mann–Whitney U and rank-biserial r:** $U = R_1 - n_1(n_1 + 1)/2$; $r = 1 - 2U/(n_1 n_2)$. Used for skewed numeric variables between two groups.
- **Spearman's ρ:** $1 - 6\Sigma d^2/[n(n^2 - 1)]$. Used for monotonic association and feature redundancy.
- **Two-proportion z test:** $z = (p_1 - p_2)/\sqrt{[P(1 - P)(1/n_1 + 1/n_2)]}$. Used for Hit@1 by phase and cancellation by match.
- **McNemar's test:** $\chi^2 = (|b - c| - 1)^2/(b + c)$, where b and c count events only one model gets right. Used to compare two rankers on the same events.
- **Paired t test and Cohen's d_z:** $t = \text{mean}(d)/(\text{SD}(d)/\sqrt{n})$; $d_z = \text{mean}(d)/\text{SD}(d)$, where d is each student's difference in Hit@1.
- **Wilson confidence interval** for a proportion, accurate near 0 and 1; used for learning curves.
- **Clustered bootstrap:** students resampled with replacement 200 times to give an interval for the AUC difference that respects dependence between a student's orders.

Appendix C: Reproducible Workflow

- Download *school_meal_synthetic_v2.zip* from release v1.0 into *data/raw/*, or regenerate it with *python src/generate_school_meal_data.py --seed 42*.
- Install the packages in *requirements.txt* (Python 3.11, pandas, scikit-learn, LightGBM, SciPy, statsmodels).
- Run *notebooks/02_eda_by_hypothesis.ipynb* for the hypothesis baselines (Figure 7).
- Run *notebooks/03_EDA_Features_Clustering.ipynb* for cleaning, EDA, features, segmentation, and the preliminary models (Tables 5–10, Figures 3–11). Intermediate results are cached, and all random processes use seed 42.